

# Lakshya JEE 2.0 (2024)

## Differential Equation

DPP-02

- The solution of  $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$  is
  - $\sin^{-1} x \sin^{-1} y = C$
  - $\sin^{-1} x = C \sin^{-1} y$
  - $\sin^{-1} x - \sin^{-1} y = C$
  - $\sin^{-1} x + \sin^{-1} y = C$
- Solution of given differential equation  $(x^3 - y^2 x^3) \cdot \frac{dy}{dx} + y^3 + x^3 y^3 = 0$  is
  - $x - \log y + C = \frac{1}{2} \left( \frac{1}{y^2} + \frac{1}{x^2} \right)$
  - $x - \log y + C = \left( \frac{1}{y^2} - \frac{1}{x^2} \right)$
  - $x + \log y + C = \frac{1}{2} \left( \frac{1}{y^2} - \frac{1}{x^2} \right)$
  - $x + \log y + C = \frac{1}{2} \left( \frac{1}{y^2} + \frac{1}{x^2} \right)$
- Find the general solution of differential equation of  $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$  is
  - $\ln \left( 1 + \sin \left( \frac{x+y}{2} \right) \right) = x + C$
  - $\ln \left( 1 + \tan \left( \frac{x+y}{2} \right) \right) = x + C$
  - $\ln \left( 1 + \cos \left( \frac{x+y}{2} \right) \right) = x + C$
  - $\ln \left( 1 + \cot \left( \frac{x+y}{2} \right) \right) = x + C$
- If solution of differential equation  $xy \frac{dy}{dx} = \frac{1+y^2}{1+x^2} (1+x+x^2)$  is
  - $\sqrt{1+x^2} = cye^{\tan^{-1} y}$
  - $\sqrt{1+x^2} = cxe^{\tan^{-1} y}$
  - $\sqrt{1+y^2} = cxe^{\tan^{-1} x}$
  - $\sqrt{1+y^2} = cye^{\tan^{-1} y}$
- Find the equation of curve through point (1, 0) and whose slope is  $\frac{y-1}{x^2+x}$ 
  - $(y-1)(x+1) + 2x = 0$
  - $2x(y-1) + x + 1 = 0$
  - $y = \frac{1-x}{1+x^2}$
  - $xy + 1 = (1+x)(1+y)$
- The solution of differential equation  $\frac{dy}{dx} = \cos(x-y)$  is
  - $y + \cot \left( \frac{x-y}{2} \right) = C$
  - $x + \cot \left( \frac{x-y}{2} \right) = C$
  - $x + \tan \left( \frac{x-y}{2} \right) = C$
  - $y + \tan \left( \frac{x-y}{2} \right) = C$
- The solution of  $\frac{dy}{dx} + 1 = e^{x+y}$  is
  - $e^{-(x+y)} + x + C = 0$
  - $e^{-(x+y)} - x + C = 0$
  - $e^{x+y} + x + C = 0$
  - $e^{x+y} - x + C = 0$

8. The differential equation of the family of curves  $y = P(x + Q)^2$  is

- (1)  $yy'' = (y')^2$   
 (2)  $2yy'' = (y')^2$   
 (3)  $2yy'' = y' + y$   
 (4)  $2yy'' = y'' - y$

9. If  $\frac{dp}{dy} = 3^{\cos y} \sin y$ , then  $p$  is equal to

- (1)  $\sin y + C$  (2)  $3^{\cos y} + C$   
 (3)  $\frac{-3^{\cos y}}{\ln 3} + C$  (4)  $3^{\sin y} + C$

10. The equation of a curve passing through the origin and satisfying the differential equation

$$\frac{dy}{dx} = (x - y)^2, \text{ is}$$

- (1)  $e^{2x} (1 - x + y) = 1 + x - y$   
 (2)  $e^{2x} (1 + x - y) = 1 - x + y$   
 (3)  $e^{2x} (1 - x + y) + (1 + x - y) = 0$   
 (4)  $e^{2x} (1 + x + y) = 1 - x + y$

11. The solution of  $\log\left(\frac{dy}{dx}\right) = ax + by$  is

- (1)  $\frac{e^{by}}{b} = \frac{e^{ax}}{a} + c$  (2)  $\frac{e^{-by}}{-b} = \frac{e^{ax}}{a} + c$   
 (3)  $\frac{e^{-by}}{a} = \frac{e^{ax}}{b} + c$  (4) None of these

12. The solution of the equation  $\frac{dy}{dx} = \frac{y^2 - y - 2}{x^2 + 2x - 3}$  is

- (1)  $\frac{1}{3} \log \left| \frac{y-2}{y+1} \right| = \frac{1}{4} \log \left| \frac{x+3}{x-1} \right| + c$   
 (2)  $\frac{1}{3} \log \left| \frac{y+1}{y-2} \right| = \frac{1}{4} \log \left| \frac{x-1}{x+3} \right| + c$   
 (3)  $4 \log \left| \frac{y-2}{y+1} \right| = 3 \log \left| \frac{x-1}{x+3} \right| + c$   
 (4) None of these



**Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.**

## **Answer Key**

1. (4)
2. (1)
3. (2)
4. (3)
5. (1)
6. (2)

7. (1)
8. (2)
9. (3)
10. (1)
11. (2)
12. (3)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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